Reflection on Constructivism

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Understandings of learning/knowledge

- Knowledge is a function of how the individual creates meaning from their own experiences
- Learning = creating meaning from experience
- Considered branch of cognitivism
    - Both conceive learning as mental activity
- Mind filters input from world to produce own unique reality
- Rationalists - mind is believed to be source of all meaning
- Empiricists - individual, direct experiences with env't considered critical
- Constructivists cross both - interaction between mind and individual
- Does not believe that knowledge is mind-independent and can be mapped onto learner
- What we know of world stems from our own interpretations of our experiences
    - Humans create meaning vs. acquiring it
    - Many meanings from any experience, cannot achieve a predetermined "correct" meaning
- Learners do not transfer knowledge from external world into memories
    - They build personal interpretations of world based on individual experiences and interactions
    - Internal representation of knowledge constantly open to change, no objective reality for learners to know
- Must examine experience to understand learning within an individual
- Learner and env'tal factors are critical
    - Specific interaction between the two that creates knowledge
- Behaviour situationally determined
    - E.g., learning of new vocab enhanced by exposure and interaction with words in context (vs. learning meaning from dictionary)
- Important that content knowledge is embedded in situation it is to be used
    - Situations co-produce knowledge and cognition through activity
- Every action is "interpretation of current situation based on entire history of previous interactions"
- Meanings of words are constantly changing - a learner's current understanding of a word and concepts will evolve with each new use
    - Important that learning happens in realistic setting and that the learning task is relevant to students' live experiences
- Impossible to isolate units of info or divide up knowledge domains according to hierarchical analysis of relationships

- Effective for stage of advanced knowledge acquisition, where initial misconceptions and biases acquired during intro stage can be discovered, negotiated and modified/removed if needed
- As learners get more knowledge, should move to constructivist approach, gives them conceptual power to deal with complex and unstructured problems

Views on memory

- Goal is for student to be able to elaborate and interpret info
- Understanding developed through continued, situated use, does not come into categorical definition that can be called on from memory
- Concept will continue to evolve with each new use as new situations, negotiations and activities recast it in a different, more densely textured form
- "Memory" is always under construction as cumulative history of interactions
- Emphasis on giving learners means to create novel and situation-specific understandings by putting together prior knowledge from different sources appropriate to current problem
    - E.g., knowledge of design activities has to be used by practitioner in too many different ways for them all to be anticipated in advance
- Emphasis on flexible use of pre-existing knowledge vs. recall of prepacked schemas
- Memory is not context-independent process
- Learning must include
    - Activity (practice)
    - Concept (knowledge)
    - Culture (context)

Views on transfer

- Transfer happens when learner is involved in authentic tasks anchored in meaningful contexts
- Context forms link with knowledge embedded in it
- Goal of instruction is to portray tasks, not define structure of learning needed to achieve the task
- If learning decontextualized, transfer probably won't happen
    - E.g., using tools, cannot just follow list of rules, need to use them in real life
- Ultimate measure of learning is based on how effective knowledge structure is in facilitating thinking and performing in system in which tools are used

Basic principles to instructional design

- Help learners explore complex topics/env'ts that move them into thinking about context like an expert user of the domain
- Learners should construct own understandings and validate them, through social negotiation
- Information from many sources important
    - E.g., do not teach new ID student facts about ID, but prepare them to use ID facts as instructional designer might
    - Performance objective not related to content, but more the process of construction
- Some strategies
    - Situating tasks in real world contexts
    - Use of cognitive apprenticeships (modeling, coaching a student toward expert performance)
    - Presentation of multiple perspectives (collab learning to dev and share diff views)
    - Examples of real "slices of life"
    - Reflective awareness
    - Giving guidance on use of constructive processes
- Principles for ID:
    - Emphasis on identification of context where skills will be learned and used
        - Anchor learning in meaningful context
    - Emphasis on learner control and capability of learner to manipulate info
        - Actively use what is learned
    - Need for info to be presented in different ways
        - Revisiting context at diff times, in rearranged contexts, for diff purposes and from diff conceptual perspectives
    - Supporting use of problem solving skills that allow learners to go "beyond info given"
        - Develop pattern-recognition skills, presenting diff ways of representing problems
    - Assessment focused on transfer of knowledge and skills
        - Presenting new problems and situations that differ from conditions of initial instruction

Formulating instruction and assessment

- Learner actively involved in process
    - But here learner is more than just an active processor of info - learner elaborates upon and interprets info
- Meaning created by learner - emphasis on learner creation
    - Learning objectives not prespecified/instruction not predesigned
- Show students how to construct knowledge, promote collaboration with others to show diff perspective can be brought to bear on a problem and to arrive at self-chosen positions, realizing basis of other views which they may disagree

- Teacher/designer role:
    - Instruct student on how to construct meaning and effectively monitor, evaluate and update constructions
    - Align and design experiences for learner so authentic relevant contexts can be experienced
- Used often in prep for lawyers, docs, architects, business men via apprenticeship and on the job training
    - Not often applied to educational area
- Student would be in an apprenticeship experience
    - E.g., novice instruction designer would work with experienced ones to learn about “needs assessment”, experts would model and coach through the authentic cause, novice would experience process in the actual context of the real world problem
        - Student would experience many additional situations, each build on and adapt to previous knowledge
        - As they gain more confidence and experience - move into collab phase, discuss with others - peers, advanced students, profs, to better articulate understanding
        - Slowly uncover what they do not know, guide them towards conceptual framing (learning)
        - Encounter complex situations, relevant books, conferences, seminars, discussions, use their knowledge to interpret situations around them
        - Moving from novice to expert

Reflection

- Learning theories seem to be a product of their time - ties to tech, methods and culture of the times
    - Behaviourism - Pavlov, Watson, Skinner early to mid 1900s
        - Built largely on animal experiments with assumption that same conditioning principles would generalize to humans
        - Fits an era focus on observable, measurable behaviour
    - Constructivist and Papert’s constructionism happened later alongside early personal computing in 1960s-70s
        - Time the valued individual explorational and self-directed thinking
        - LOGO and “turtle” example of this
- Are we really discovering better theories of learning over time, or is each generation just building theories that match the tools of its time?
    - E.g., Trojan Mouse idea - tech is sometimes just a vehicle for moving in a particular philosophy of learning
        - Suggests theory and tech are not neutral or separate
        - They shape each other?
- Therefore, is the current trendy learning approach “right” or “better”?
    - E.g., gamification, AI-assisted

- Or does it just fit current technology era we are in?